👑 CODERKING STUDIO

Official Technical Roadmaps & Architecture Guides

90-DAY MASTERCLASS

90 Days of Linux Internals & Systems Programming

Complete 90-day syllabus: Syscalls & Ring 0 ➔ Virtual Memory ➔ VFS & Ext4 ➔ Process Scheduling ➔ eBPF Kernels.

🐧 Month 1: User to Kernel Transition (Day 1-30)

- **Day 1-10:** CPU protection rings (Ring 0 vs 3), Syscall dispatch tables.
- **Day 11-20:** Process lifecycle: `fork()`, `clone()`, `execve()`, COW pages.
- **Day 21-30:** Deep dive with `strace`, `ltrace`, and `/proc` filesystem.

💾 Month 2: Memory & Filesystems (Day 31-60)

- **Day 31-40:** Virtual memory, MMU, Page tables, TLB, Page faults.
- **Day 41-50:** Virtual File System (VFS), Inodes, Dentries, Ext4 disk layout.
- **Day 51-60:** Linux page cache, Dirty pages, `fsync()` and zero-copy.

⚡ Month 3: Concurrency & eBPF (Day 61-90)

- **Day 61-70:** Completely Fair Scheduler (CFS), Context switching cost.
- **Day 71-80:** Linux networking stack: Socket buffers (sk_buff), Epoll internals.
- **Day 81-90:** Writing in-kernel eBPF programs, XDP packet filtering.

💻 Inspect System Calls Live

Trace any Linux binary without source code:

```
# Trace all syscalls made by 'ls'
strace -c ls -la

# Inspect open file descriptors
ls -l /proc/$$/fd
```

Follow @coderking__1 for daily kernel teardowns!

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Official GitHub: **github.com/shubh74/coderking-roadmaps**